

EDUCATION

Department of Automation, Tsinghua University <i>Ph.D. in Control Science and Engineering</i>	Beijing, China 2022 - 2028 (<i>expected</i>)
• Advisor: Prof. Shiji Song • Research area: Machine Learning (including LLM) for Operations Research	
Department of Precision Instrument, Tsinghua University <i>B.E. in Measurement and Control Technology and Instrument</i>	Beijing, China 2018 - 2022
• GPA: 3.66/4.00, Rank: 8/67	
School of Economics Management, Tsinghua University <i>Minor in Economics and Finance</i>	Beijing, China 2019 - 2022

PUBLICATIONS AND WORKING PAPERS

1. **Haofeng Yuan**, Jianing Peng, Jieyi Bi, Ni Zhang, Shiji Song, Zhiguang Cao. FormuEvo: LLM-Guided Evolution for Discovering Solver-Efficient Mixed-Integer Programming Formulations. *The Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
2. **Haofeng Yuan**, Rongping Zhu, Wanlu Yang, Shiji Song, Keyou You, Wei Fan, C. L. Philip Chen. Deep Reinforcement Learning for Traveling Purchaser Problems. *IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI)*, 2025.
3. **Haofeng Yuan**, Lichang Fang, Shiji Song. A Reinforcement-Learning-Based Multiple-Column Selection Strategy for Column Generation. *AAAI Conference on Artificial Intelligence (AAAI)*, 2024.
4. **Haofeng Yuan**, Peng Jiang, Shiji Song. The Neural-Prediction Based Acceleration Algorithm of Column Generation for Graph-Based Set Covering Problems. *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2022.
5. Wanlu Yang, Linyu Liu, **Haofeng Yuan**, Shiji Song. Unified Scheduling Model for High-Speed Train Timetable Optimization and Rescheduling Based on Deep Reinforcement Learning. *IEEE Transactions on Intelligent Transportation Systems (TITS)*, 2025.
6. Lichang Fang, **Haofeng Yuan**, Bokui Chen, Shiji Song. Learning to Stabilize Column Generation. *International Joint Conference on Neural Networks (IJCNN)*, 2025.
7. Lichang Fang, **Haofeng Yuan**, Yuli Zhang, Shiji Song. Accelerating Column Generation Algorithm Using Machine Learning Based Column Elimination. *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2023.
8. **Haofeng Yuan**, Zhiguang Cao, Jianing Peng, Wanlu Yang, Yuli Zhang, Shiji Song. Dynamic Scheduling for Hot Rolling Production by Hierarchical Reinforcement Learning and Heterogeneous Graph Neural Networks. Under minor revision at *IEEE Transactions on Industrial Informatics (TII)*, 2026.
9. Ni Zhang#, **Haofeng Yuan**#, Jianan Zhou, Shiji Song, Zhiguang Cao. LLM as Exogenous World Model for Dynamic Scheduling. Under review at *AAAI Conference on Artificial Intelligence (AAAI)*, 2027. (#co-first author)
10. Wanlu Yang, **Haofeng Yuan**, Jianing Peng, Shiji Song. Graph-Based Meta-Reinforcement Learning for Passenger-Oriented Train Timetable Rescheduling in a High-Speed Railway Network. Under major revision at *IEEE Transactions on Intelligent Transportation Systems (TITS)*, 2026.
11. Jianing Peng, Yuli Zhang, **Haofeng Yuan**, Wanlu Yang, Shiji Song. Logic-Based Benders Decomposition for Production Planning with State Transition-Based Ramp-Up. Under major revision at *Omega*, 2026.

RESEARCH PROJECTS	AI for Social Sciences Research Award Program	2025.01 - 2026.12
	<i>School of Social Science, Tsinghua University</i>	
	Joint Optimization of Battery Swapping and Charging Scheduling for Electric Vehicles <i>Tsinghua University & SANY Group Ltd.</i>	2023.02 - 2023.06
	Intelligent Online Optimization Method for High-Speed Railway Scheduling <i>National Natural Science Foundation of China</i>	2021.09 - 2024.12
INTERNSHIPS	Singapore Management University Singapore, Singapore	2025.08 - 2026.08
	• Visiting Ph.D. student advised by Prof. Zhiguang Cao. • Researched on large language models (LLM) in optimization and scheduling.	
	Sinotrans (Angola) Ltd. Luanda, Angola	2024.07 - 2024.08
	• Investigated in the operation and management of Chinese enterprises in Africa. • Worked on robust truck fleet scheduling and routing optimization algorithm.	
	Huawei Technologies Ltd. Beijing, China	2022.04 - 2022.07
	• Worked on efficient pedestrian detection and attribute recognition algorithm.	
	Commercial Aircraft Corporation of China Ltd. Shanghai, China	2021.07 - 2021.08
	• Participated in the development of real-time flight data monitoring system.	
	Graph Origin Ltd. Beijing, China	2021.05 - 2021.07
	• Worked on generative models for 3D skeleton retargeting and style transfer. • Designed a solution approach for unnatural torsion of joints in motion capture.	
	Government of Liangshan Prefecture Sichuan, China	2019.07 - 2019.08
	• Investigated in Yi culture and targeted poverty alleviation in Liangshan prefecture.	
AWARDS AND HONORS	• Comprehensive Excellence Scholarship, Tsinghua University	2019, 2023, 2024, 2025
	• Academic Excellence Scholarship, Tsinghua University	2020
	• Golden Prize Team for Social Practice, Tsinghua University	2019
	• First Prize in China Undergraduate Physics Tournament	2019
SKILLS	Languages: CET-4, CET-6.	
	Programming: Python, C/C++, MATLAB, Stata.	
ACADEMIC SERVICES	Reviewer for: <i>INFORMS Journal on Computing (IJOC),</i>	
	<i>IEEE Transactions on Intelligent Transportation Systems (TITS),</i>	
	<i>IEEE Transactions on Industrial Informatics (TII),</i>	
	<i>IEEE Transactions on Automation Science and Engineering (TASE),</i>	
	<i>IEEE Transactions on Systems, Man and Cybernetics: Systems (TSMC-A),</i>	
	<i>Transportation Research Part E (TR-E),</i>	
	<i>Advances in Neural Information Processing Systems (NeurIPS),</i>	
	<i>Empirical Methods in Natural Language Processing (EMNLP),</i>	
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